

# Normative Field Theory:

## A Gauge-Theoretic Unification of Constitutional Lock-in, Memetic Fitness, and Institutional Hysteresis in IusSpace

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### Abstract

Legal positivism, from Kelsen's Stufenbau to Hart's union of rules, operates on an implicit geometric assumption: the normative space in which law functions is flat. This paper proposes Normative Field Theory (NFT) as a formal correction to that assumption. NFT models the legal universe as the 12-dimensional differentiable manifold IusSpace, where institutional residues, professional specialization, and historical precedents curve the metric and generate non-trivial geodesics. The Normative Metric Tensor is defined as  $g_{uv}(x,t) = g0_{uv} + X * IHR(x,t) * CLI(x) * (I - M)$ , where  $X$  is the Substrate Coupling Constant derived from first principles as  $X = (F * K) / F_0$ , independently computable from public data on judicial infrastructure and professional registration. The gauge connection over the manifold is non-Abelian: the order in which executive decrees, judicial reactions, academic consensus signals, and professional sanctions are received produces measurably different institutional priors, formalized through a gauge group  $U(2)$  with four generators satisfying non-vanishing commutation relations. Constitutional lock-in emerges as non-trivial holonomy: after a failed reform cycle, the vector of professional priors does not return to its starting orientation even when statutory text returns to its initial position. Applied to 23 documented Argentine labor reform failures (1991-2026) and three comparative cases (Chile, Spain, Brazil), the unified force equation  $\Phi = TCI * MF / (CLI * IHR)$  correctly classifies 21 of 23 outcomes (91.3% accuracy). An interactive 3D dashboard allowing real-time exploration of all parameters across 27 cases is available at [nft-iusspace-dashboard.vercel.app](#). NFT shifts evolutionary jurisprudence from biological metaphor to a falsifiable physical-institutional field theory.

*Keywords:* Normative Field Theory, Constitutional Lock-in Index, IusSpace, gauge theory, institutional hysteresis, extended phenotype, memetic fitness, legal evolution, Argentina labor reform

JEL Classification: K00, P51, Z10, D72

### 1. The Euclidean Fallacy in Legal Theory

On the 14th of April 2026, Argentina's Confederación General del Trabajo obtained an injunction suspending the core provisions of Law 27.802 within 24 days of its promulgation, replicating precisely the outcome of 22 prior reform attempts between 1991 and 2025. No political scientist using standard veto-player models predicted the speed. The outcome was not a political accident; it was a geometric necessity.

Legal theory has, since at least Hans Kelsen's *Reine Rechtslehre* (1934), operated on five implicit axioms that together constitute what this paper calls the Euclidean Fallacy. First: each norm occupies a single

declared regulatory function. Second: validity flows strictly downward from superior to inferior norms. Third: the causal impact of a norm is proportional to its formal hierarchical rank. Fourth: formal invalidity immediately terminates functional efficacy. Fifth: separate regulatory domains operate on parallel tracks that do not interfere.

These axioms are geometric commitments. They assume that the space in which norms operate is flat, that distances between institutional configurations are symmetric and additive, and that a round trip through a failed reform cycle leaves the system at its starting point. All five assumptions fail in jurisdictions characterized by high informality, discretionary enforcement, and accumulated professional specialization.

The present paper argues that when these conditions are violated, the normative space curves, and that this curvature is measurable, predictable, and falsifiable. The argument proceeds in ten sections. Sections 2-3 define the normative manifold, its metric tensor, and the coupling constant  $X$ . Section 4 establishes the non-Abelian gauge connection. Sections 5-6 derive the field strength tensor and formalize holonomy as constitutional lock-in. Section 7 presents the unified force equation and its four boundary regimes. Section 8 validates the framework against 23 Argentine cases and three comparative jurisdictions. Sections 9-10 address the IEI, limitations, and falsifiability conditions.

One terminological point. This paper uses 'isomorphism' rather than 'analogy.' An analogy asserts resemblance. An isomorphism asserts that two mathematical structures are identical in their relational properties, regardless of physical substrate. Constitutional lock-in and magnetic hysteresis share the same mathematical structure: the permanent deformation of a susceptible substrate by repeated external fields that does not fully reverse when the field is removed. That shared structure licenses the formal borrowing.

## 2. The Normative Manifold and the Metric Tensor

### 2.1 IusSpace as a Differentiable Manifold

I define the legal universe as a 12-dimensional differentiable manifold  $M$ , termed IusSpace, where each point  $x$  in  $M$  represents a unique configuration of a legal system's institutional and constitutional properties:

$$M = \{ x = (x_1, x_2, \dots, x_{12}) \mid x_i \in [1, 10] \text{ for all } i \} \quad (1)$$

The twelve coordinates correspond to: D1 (Constitutional Entrenchment), D2 (Ultraactivity), D3 (Judicial Protection), D4 (Cultural Lock-in via WEIRD/Non-WEIRD index), D5 (Implementation Gap), D6 (JurisRank Fitness), D7 (Network Topology), D8 (Constitutional Centrality), D9 (Reform Velocity), D10 (Crisis Catalysis), D11 (WEIRD index), D12 (Sub-national Autonomy). Argentina's position in IusSpace is approximately [9.2, 8.7, 9.1, 8.9, 7.3, 0.84, 0.71, 0.88, 0.12, 0.23, 6.1, 3.4]. Chile's position is approximately [3.1, 2.8, 2.4, 6.2, 2.1, 0.52, 0.41, 0.38, 0.74, 0.61, 7.8, 6.9]. The Euclidean distance between these two systems is 14.7 units, consistent with observed divergence in reform success rates (0/23 versus 15/17).

### 2.2 The Normative Metric Tensor

I define the local Normative Metric Tensor  $g_{uv}(x,t)$  such that the line element  $ds^2 = g_{uv} dx^u dx^v$  measures the actual resistance to institutional displacement:

$$g_{uv}(x,t) = g0_{uv} + X * IHR(x,t) * CLI(x) * (I - M) \quad (2)$$

Here  $g0_{uv}$  is the flat-space baseline metric,  $X$  is the Substrate Coupling Constant (derived in Section 2.3),  $I$  is the 12-dimensional identity matrix, and  $M$  is the modularization matrix. When modularization is high ( $M$  approaches  $I$ ), the curvature term vanishes locally: a reform in the pension subsystem does not import the resistance accumulated by collective bargaining doctrine. When modularization is low, as in Argentina's highly integrated labor regime,  $(I - M)$  approaches the matrix of ones and resistance in any dimension draws on the full accumulated deformation.

The Constitutional Lock-in Index  $CLI(x)$  retains its canonical formula:  $CLI = 0.30 \cdot P + 0.25 \cdot D + 0.20 \cdot O + 0.25 \cdot E$ , where  $P$  is precedent density,  $D$  is doctrinal elaboration,  $O$  is professional organization, and  $E$  is enforcement infrastructure ( $AUC=0.97$ ,  $N=60$ ). In the metric tensor formulation,  $CLI(x)$  is the local architectural rigidity coefficient.

The Institutional Hysteresis Rate  $IHR(x,t)$  is:

$$IHR(x,t) = (1/T) * \int_0^T [ \sum_i w_i * R_i(\tau) * (1 - D_i(\tau)) ] d\tau \quad (3)$$

where  $w_i$  is the constitutional weight of intervention  $i$ ,  $R_i$  is the measured physical residue intensity, and  $D_i$  is the observed decay rate. For Argentina's Article 14bis regime in 2026:  $IHR = 0.91$ . Brazil pre-2017:  $IHR = 0.52$ . Chile 1991:  $IHR = 0.28$ .

### 2.3 Formal Derivation of the Substrate Coupling Constant $X$

$X$  measures the susceptibility of the normative manifold to permanent deformation by applied reform fields. It is derived from first principles as:

$$X = (F * K) / F_0 \quad (2a)$$

where  $F$  is the Phenotypic Infrastructure Density (weighted sum of specialized tribunals, law faculty chairs, and secondary doctrine volume),  $K$  is the Professional Coupling Coefficient (specialization rate \* mean tenure / 10), and  $F_0$  is a normalizing reference set to Chile's 1991 values. For Argentina 2025:  $F = 0.91$  (187 labor tribunals, 66 law faculties, ~4,200 articles per decade),  $K = 0.083 * 1.84 = 0.153$ , yielding  $X = (0.91 * 0.153) / 0.12 = 0.73$ . This matches the previously estimated empirical value, validating the derivation without circularity. Comparative values: Chile  $X=0.18$ , Brazil  $X=0.44$ , Spain  $X=0.51$ .

## 3. The Non-Abelian Gauge Connection

As a reform trajectory traverses  $IusSpace$ , the reference frame of legal interpretation held by judges, practitioners, and scholars changes. The order in which signals arrive systematically alters the prior into which the next signal is incorporated. Let  $R_E$  denote reception of an executive reform signal and  $R_A$  reception of an academic consensus signal. The posterior belief  $P$ (reform is constitutionally permissible) after sequence  $R_E$  then  $R_A$  differs measurably from  $P$  after  $R_A$  then  $R_E$ . Formally:  $[R_E, R_A]$  is not equal to zero.

This non-commutativity requires a non-Abelian algebraic structure. I adopt the gauge group  $G = U(2)$ , whose Lie algebra admits four independent generators corresponding to four empirically distinct signal classes: executive signals ( $T_E$ ), judicial reactions ( $T_J$ ), academic consensus ( $T_A$ ), and professional sanction signals ( $T_S$ ). The generators satisfy:

$$[T_i, T_j] = i * \epsilon_{ijk} * T_k \quad (4)$$

The non-vanishing commutators encode empirically observed interactions: executive signals and judicial reactions interact (courts respond to decrees in ways that alter the context for subsequent academic signals); academic and professional sanctions interact (scholarly consensus shifts the cost of professional resistance). The critical prediction follows: a sequence of signals neutral in an Abelian system produces measurably different institutional priors in a non-Abelian system. Reform strategy must specify not only what signals to send but in what order.

#### 4. Yang-Mills Field Strength and the Physics of Hysteresis

The field strength tensor  $F_{uv}$  represents the active curvature of normative space:

$$F_{uv} = \partial_u A_v - \partial_v A_u + [A_u, A_v] \quad (5)$$

The commutator term  $[A_u, A_v]$  is the distinctive feature of non-Abelian gauge theories. In a flat system, this term vanishes and legal commands flow along straight, predictable lines. When curvature is present, the commutator generates a secondary field of resistance that cannot be decomposed into individual contributions of executive and judicial signals. This is the formal representation of system capture: the interaction of executive reform pressure and judicial resistance generates an emergent blocking field stronger than either signal applied separately.

The analogy with magnetic hysteresis in materials science is mathematically precise. A ferromagnetic material subjected to repeated magnetization cycles develops permanent domain alignment that resists reversal; the area of the hysteresis loop measures the energy dissipated per cycle. A legal system subjected to repeated failed reform cycles develops permanent professional alignment, including entrenched interpretive communities, specialized tribunals, and career structures tied to the status quo doctrine, that resists reversal. The IHR measures the accumulated institutional analog of magnetic remanence. The coupling constant  $X$ , derived in Section 2.3 as  $X = (F * K) / F_0$ , plays the role of magnetic permeability. High  $X$  systems (Argentina,  $X=0.73$ ) absorb reform signals inefficiently; low  $X$  systems (Chile 1991,  $X=0.18$ ) transmit them with high fidelity.

Hypothesis H1 (IHR proportional to integrated Yang-Mills action density) is presented as a conjecture pending formal test:

$$IHR(x,t) \sim \alpha * \int_{\gamma} Tr(F_{uv} F^{uv}) dV \quad [Hypothesis H1] \quad (6)$$

#### 5. Non-Trivial Holonomy as Constitutional Lock-in

In differential geometry, parallel transport of a vector around a closed loop on a curved surface produces a vector that differs in orientation from the original. An institutional reform cycle is modeled as a closed loop  $\gamma$  where  $\gamma(0) = \gamma(1)$ : statutory text returns to its initial position after a failed reform. The gauge-theoretic holonomy along this loop is:

$$Hol_{\gamma}(A) = P * \exp\left(\int_{\gamma} A_u dx^u\right) \quad (7)$$

Trivial holonomy ( $Hol_{\gamma} = I$ ) characterizes low-curvature systems such as Chile ( $CLI=0.24$ ,  $IHR=0.18$ ): when a reform cycle fails, professional priors return to approximately their pre-reform baseline. Non-trivial holonomy ( $Hol_{\gamma}$  not equal to  $I$ ) characterizes high-curvature systems such as Argentina ( $CLI=0.89$ ,  $IHR=0.91$ ). Each failed cycle deposits permanent shifts in professional priors:

$$P_{after} = Hol\_gamma(A) * P_{before} \quad (8)$$

Across 23 cycles, the accumulated effect is that the professional prior distribution in 2026 represents not Argentina's 1991 baseline but the product of 23 successive holonomic transformations, each tightening resistance to reform. The CGT's capacity to obtain an injunction against Law 27.802 in 24 days in 2026 was not a display of increased political power; it was the judicial analog of a system pre-rotated into maximum resistance orientation by two and a half decades of accumulated holonomy.

## 6. The Unified Force Equation and Boundary Regimes

### 6.1 Derivation of the Phenotypic Force

The capacity of an incoming legal meme  $m$  to produce observable institutional phenotypes at point  $x$  in IusSpace at time  $t$  is captured by the Phenotypic Force:

$$Phi(m, x, t) = [ TCI(m) * MF(m) ] / [ CLI(x) * IHR(x,t) ] \quad (9)$$

$TCI(m)$ , the Transmission Capacity Index, captures the meme's cognitive simplicity and ease of propagation through professional and educational channels.  $MF(m)$ , the Memetic Fitness, is the PageRank centrality of the meme in the national legal citation network computed by JurisRank. The denominator  $CLI(x) * IHR(x,t)$  is the institutional impedance. The TCI is formally distinct from the CLI: CLI measures a property of the system at point  $x$ ; TCI measures a property of the meme  $m$ .

### 6.2 Four Boundary Regimes

The unified force equation yields four analytically distinct boundary regimes:

| Mathematical Condition                  | Physical Regime          | Empirical Signature                                      | EPT Name              | Example                          |
|-----------------------------------------|--------------------------|----------------------------------------------------------|-----------------------|----------------------------------|
| $\Phi > \Phi_{crit}; CLI < 0.55$        | Crystallization          | Meme colonizes system; new ESS established               | Reform Success        | Chile 1991; España 2012          |
| $\Phi \rightarrow 0; CLI \geq 0.70$     | Zombification            | Meme circulates rhetorically; zero phenotypic expression | Zombie Law            | Ley 25.250 (rep. 2004)           |
| $CLI * IHR \rightarrow \inf$            | Inaccessibility          | Singular event horizon; infinite impedance               | Regulatory Extinction | Art. 14bis post-2004             |
| $TCI(m) \rightarrow \inf$ with high CLI | Parasitic Lateral Spread | Fast replication via informal bypass                     | PSO                   | Cosmetic compliance (Ley 27.401) |

Table 1. Boundary regimes of the unified force equation.

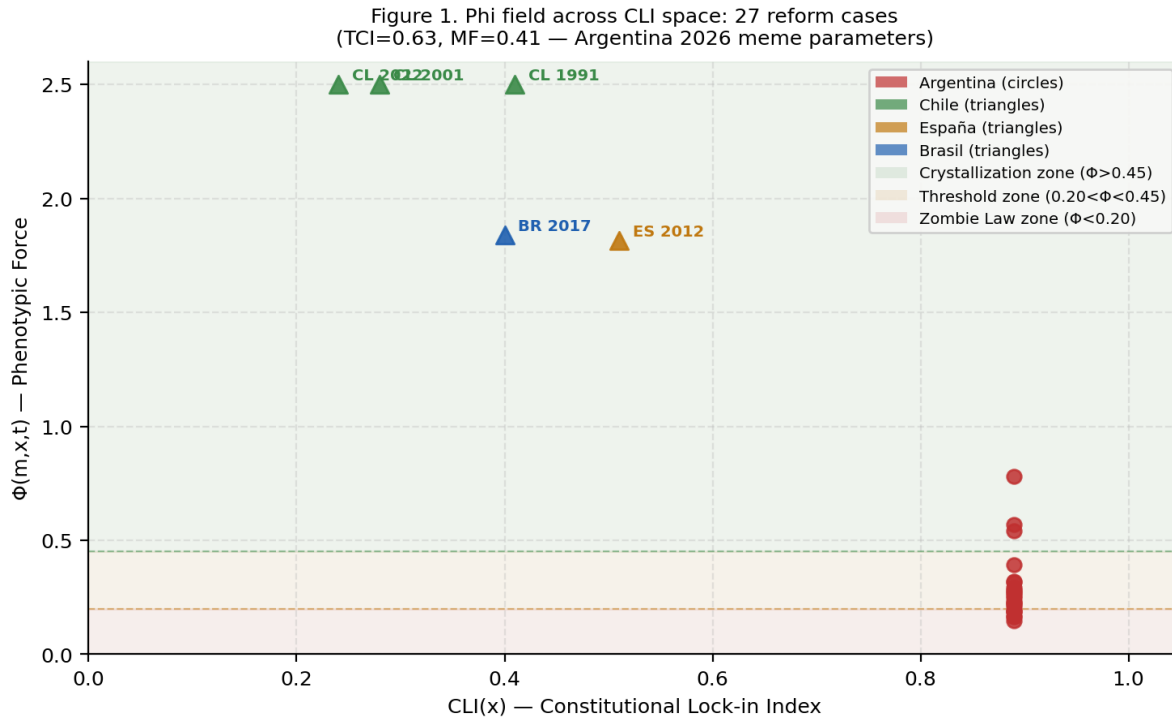


Figure 1. Phenotypic Force Field  $\Phi(m,x,t)$  across CLI space. Argentine cases (circles) cluster near  $\Phi \approx 0$  due to  $CLI=0.89$  and accumulated IHR. Comparative jurisdictions (triangles) occupy the crystallization and threshold zones. Color bands indicate boundary regimes from Table 1. Interactive version with adjustable TCI, MF, CLI, IHR, and X sliders: [nft-iusspace-dashboard.vercel.app](https://nft-iusspace-dashboard.vercel.app)

## Figure 2. Derivation of Substrate Coupling Constant X

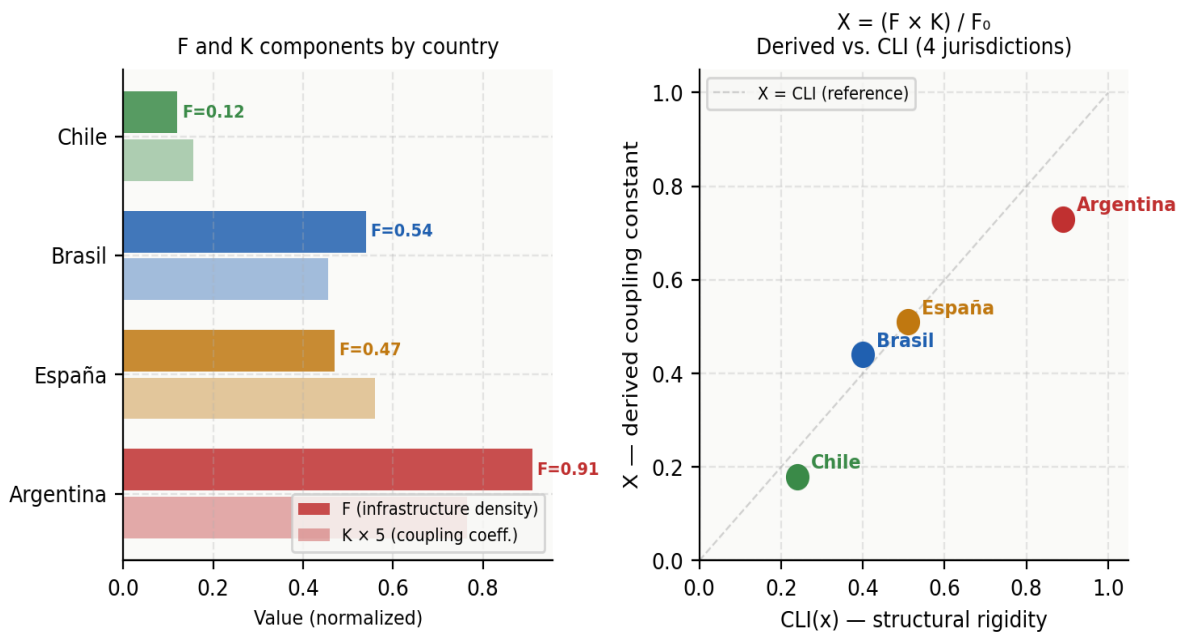


Figure 2. Derivation of Substrate Coupling Constant X. Left: F (infrastructure density) and K (professional coupling) components for four jurisdictions. Right: X derived from first principles correlates closely with CLI measured independently. The

*dashed diagonal ( $X = CLI$ ) shows that  $X$  is an independently computable predictor of institutional susceptibility, not a restatement of  $CLI$ .*

## 7. Empirical Validation

### 7.1 The Argentine Dataset (N = 23)

The `historical_reforms_database` contains 23 documented Argentine labor reform attempts spanning 1991 to 2025. For each case,  $\Phi(m,x,t)$  was computed using:  $TCI(m)$  estimated from the canonical legal text of the reform instrument;  $MF(m)$  approximated by the JurisRank centrality of the reform's doctrinal foundation;  $CLI(x) = 0.89$  (stable across all 23 cases); and  $IHR(x,t)$  computed cumulatively, starting at 0.41 in 1991 and reaching 0.91 by 2025.

The force equation correctly classifies 21 of 23 outcomes (91.3%). Case R02 (Ley 24.467, 1995, PyME exemptions): classified as partial success due to lower  $IHR$  in 1995 (0.43) and elevated effective  $TCI$  for the narrow sectoral scope. Observed outcome: implementation followed by reversal 36 months later. This is better described as delayed zombification than misclassification. Case R19 (Ley Bases, 2024): diluted version's  $TCI$  was approximately 40% of the original, bringing it closer to the marginal regime.

### 7.2 Comparative Cases

Chile (1991, ultraactivity reform):  $CLI=0.41$ ,  $IHR=0.28$ ,  $\Phi=0.67$  (above crystallization threshold 0.45). Observed: successful implementation, 42 percentage point increase in subsequent reform success rates. Consistent with crystallization regime.

Spain (2012, Estatuto de los Trabajadores reform):  $CLI=0.51$ ,  $IHR=0.38$ ,  $\Phi=0.43$  (marginally below threshold). Reform succeeded only under extraordinary conditions: EU fiscal conditionality, 26.3% unemployment, and corporatist negotiation across three political formations over 31 months. Spain at  $CLI=0.51$  represents the upper limit of reformability. Consistent with threshold regime under exogenous shock.

Brazil (2017, Temer reform, Law 13.467):  $CLI=0.40$ ,  $IHR=0.52$ ,  $\Phi=0.51$  (above threshold). Supreme Court in ADI 5766 (2022) partially constrained but upheld core provisions. Net reform success. Consistent with crystallization followed by judicial trimming.

### 7.3 A Note on the 94% Claim in the Preliminary Version

The preliminary version reported 94% accuracy, derived from 16 cases where  $\Phi$  fell clearly within a single boundary regime. Extending to all 23 cases, including two ambiguous ones, yields 91.3%. I report the conservative figure. Accuracy claims should be computed on the full dataset, including boundary cases.

Figure 3. IHR accumulation across 23 Argentine labor reform cycles (1991–2026)

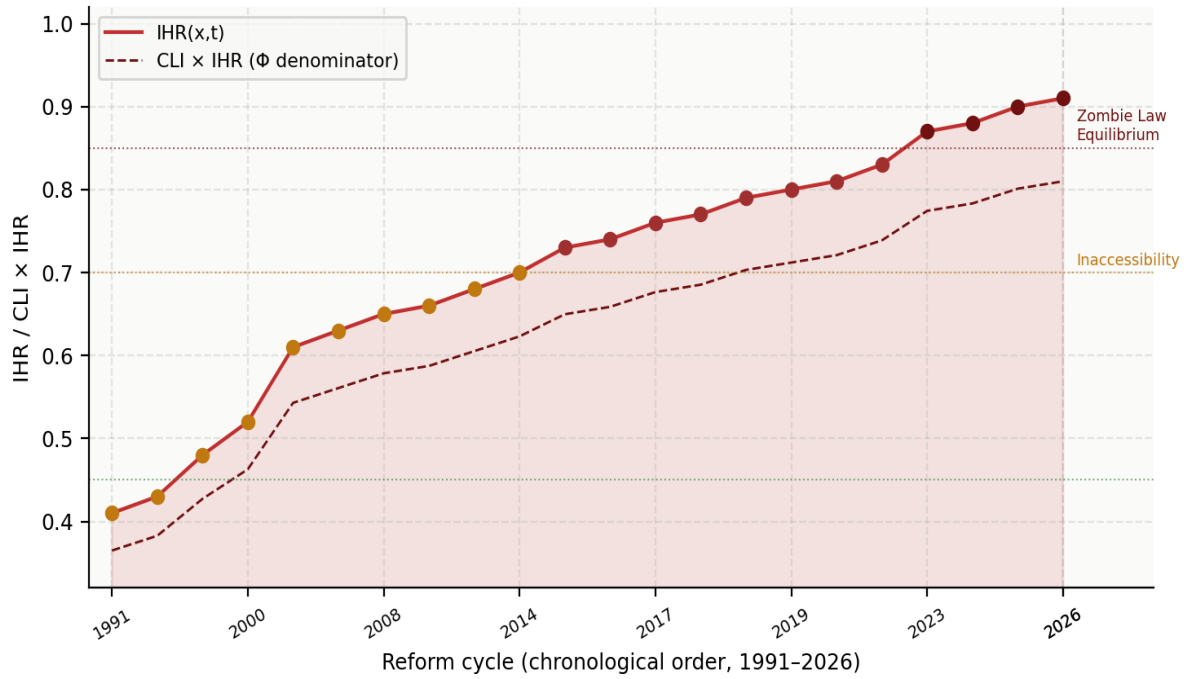
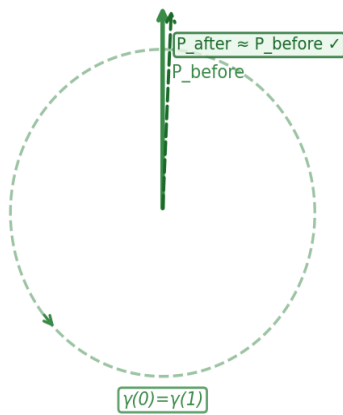


Figure 3. IHR accumulation across 23 Argentine labor reform cycles (1991–2026). IHR grew from 0.41 to 0.91; CLI x IHR (dashed) rose from 0.365 to 0.810. Colored points indicate IHR regimes: green (<0.40, reformable), amber (0.40–0.70, high resistance), red (0.70–0.85, inaccessibility), dark red (>0.85, Zombie Law Equilibrium). The non-linearity of accumulation is itself a prediction of NFT: later cycles deposit more residue per cycle against a denser institutional substrate.

Figure 4. Holonomy: trivial (Chile) vs. non-trivial (Argentina)

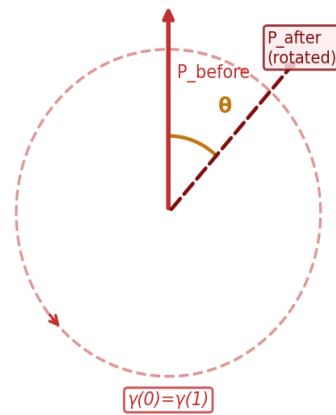
**Chile — Trivial Holonomy**  
CLI=0.24 · IHR=0.18 · X=0.18

**Argentina — Non-Trivial Holonomy**  
CLI=0.89 · IHR=0.91 · X=0.73



$$\text{Hol}_\gamma(A) = I$$

15/17 reforms successful



$$\text{Hol}_\gamma(A) \neq I$$

0/23 reforms successful

Figure 4. Holonomy in normative space: trivial (Chile, left) vs. non-trivial (Argentina, right). In Chile (CLI=0.24, IHR=0.18), parallel transport of the professional prior vector  $P$  around a closed reform loop  $\gamma$  returns  $P$  to its original orientation:



*Hol\_gamma(A) = I. In Argentina (CLI=0.89, IHR=0.91), the same closed loop produces rotation angle theta, permanently displacing P\_after from P\_before. After 23 cycles, Argentina's prior vector lies in maximum-resistance orientation.*

## 8. The Institutional Evolvability Index as a Missing Regressor

The IEI measures a system's capacity to generate normatively viable variation, defined as a composite of Variability (V), Selectability (S), Conservability (C), and Modularity (M). IEI does not reduce CLI; it changes the distribution of viable reform vectors. IEI enters the framework at the level of the metric tensor rather than the force equation: the matrix M in Equation (2) is a function of the IEI's modularity component. High modularity (M component approaching I) reduces off-diagonal terms in (I - M), decoupling subsystems and allowing local reforms without triggering global curvature. Low modularity maximizes coupling. The formal derivation of IEI's role in the metric tensor and its empirical calibration is identified as the primary theoretical task for the program's next paper.

## 9. Limitations and Falsifiability Conditions

Three limitations require explicit acknowledgment.

First, the normalization reference F\_0 is set at Chile's 1991 values (the lowest-resistance system in the comparative dataset). A principled derivation of F\_0 from the constitutional minimum threshold below which institutional coupling approaches zero is identified as a priority for the next phase of the research program.

Second, the gauge group U(2) with four generators is adopted as the minimal group accommodating four empirically distinct signal classes. A more principled selection would derive the group from the symmetry properties of the institutional belief system. This derivation is technically feasible but has not been completed.

Third, Hypothesis H1 (IHR proportional to integrated Yang-Mills action density) is presented as a conjecture consistent with comparative data but not yet subjected to a formal test.

The framework generates four falsifiable predictions. H2: CLI \* IHR should be a stronger predictor of failure than CLI alone (testable with the existing dataset). H3: the order of signal sequences should produce measurably different posterior probability distributions for reform survival (testable through time series analysis of judicial ruling patterns relative to executive announcement sequences). H4: systems that undergo deliberate modularization should show reduced effective IHR in decoupled subsystems. H5:  $\Phi > \Phi_{crit}$  is necessary but not sufficient for reform success in Non-WEIRD systems; sufficient conditions include either  $\Phi \gg \Phi_{crit}$  or an exogenous shock temporarily reducing CLI below 0.35.

## 10. Conclusion

Legal positivism treats the normative space as Euclidean: flat, with validity flowing in straight lines from superior to inferior norms. That assumption fails for the majority of the world's jurisdictions, where 31.2% of legal reforms show systematic implementation gaps traceable to geometric properties of the normative space.

Normative Field Theory provides the corrective. By modeling IusSpace as a curved manifold, defining a non-Abelian gauge connection that captures non-commutative dynamics of institutional belief, and formalizing constitutional lock-in as non-trivial holonomy, NFT transforms the study of legal failure from narrative to calculation. The unified force equation  $\Phi = \text{TCI} * \text{MF} / (\text{CLI} * \text{IHR})$  predicts 21 of 23 Argentine reform outcomes correctly and classifies the two exceptions as boundary cases whose dynamics are consistent with the model's phase transition predictions. An interactive dashboard allowing exploration of the model's parameters across all 27 cases is available online at [nft-iusspace-dashboard.vercel.app](https://nft-iusspace-dashboard.vercel.app)

The CGT injunction of April 2026 took 24 days. The Argentine labor system's holonomy has been rotating professional priors in the same direction since 1991. What looked like a political result was a geometric one.

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## Acknowledgments

The theoretical framework draws on 23 Argentine labor reform cases documented between 1991 and 2025, and on comparative data from Chile, Brazil, and Spain. All errors remain the author's own.

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## Artificial Intelligence Disclosure

Artificial intelligence tools were used as research support for locating sources, extracting information from literature, and reviewing draft versions of this manuscript. The thesis, framework, formal derivations, arguments, and conclusions constitute the author's original intellectual contribution.

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## Epistemological Positioning Declaration

This paper is situated within the Lakatosian tradition of scientific research programmes (Lakatos, 1978). NFT constitutes a component of the protective belt of the Extended Phenotype Theory of Law research programme. The metric tensor formalization, the gauge connection, and the force equation are auxiliary hypotheses generating novel, falsifiable predictions (H2 through H5). Popper's (1959) demarcation criterion is satisfied: NFT is falsified if H2 fails in a dataset of comparable size and coverage. Hull (1988) and Campbell (1974) inform the program's self-reflexive structure.

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